

EXPERIMENT 6

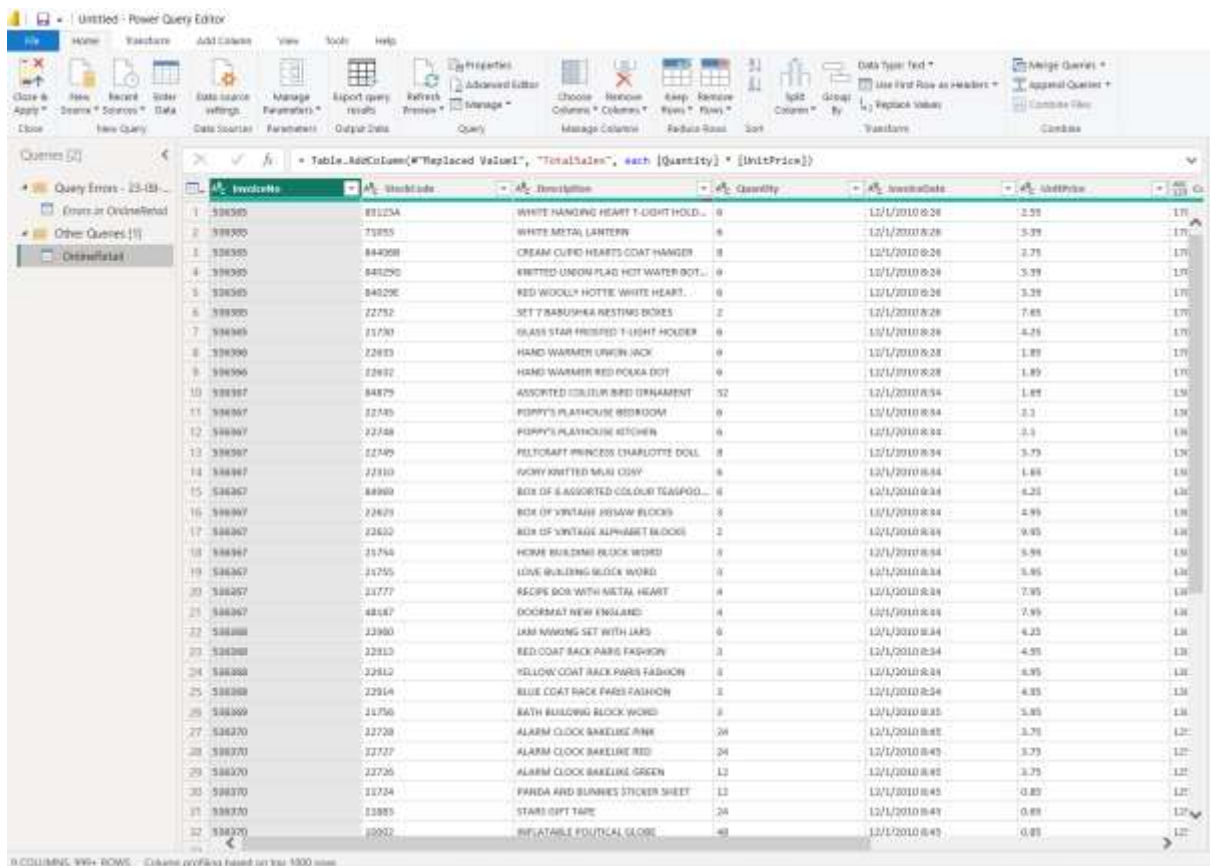
DATA VISUALIZATION USING POWER BI

Aim

To import, transform, analyze, and visualize data from OnlineRetail.csv using Power BI Desktop and create interactive dashboards and reports for decision-making

Algorithm 1: Importing Data into Power BI

- **Step 1:** Open Power BI Desktop.
- **Step 2:** Select **Home** → **Get Data**.
- **Step 3:** Choose **Text/CSV**.
- **Step 4:** Browse and select OnlineRetail.csv.
- **Step 5:** In the preview window, click **Transform Data** to open Power Query Editor.



The screenshot displays the Power Query Editor interface. The main area shows a table with the following columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, and UnitPrice. The data is filtered to show the first 32 rows. The table contains various items such as 'WHITE HANGING HEART T-LIGHT HOLDER', 'WHITE METAL LANTERN', 'CREAM CUPID HEARTS COAT HANGER', and 'RED WOOLLY HOTTE WHITE HEART'. The status bar at the bottom indicates '9 COLUMNS, 996+ ROWS'.

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice
590305	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:28	2.55
590305	73855	WHITE METAL LANTERN	6	12/1/2010 8:28	3.39
590305	844068	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:28	2.75
590305	840290	KWIFFED UNION FLAG HOT WATER BOTT...	6	12/1/2010 8:28	5.39
590305	840290	RED WOOLLY HOTTE WHITE HEART	6	12/1/2010 8:28	5.39
590305	22752	SET 7 BARBUSHEA RESTING BONES	2	12/1/2010 8:28	7.65
590305	11730	GLASS STAR FRAMED T-LIGHT HOLDER	6	12/1/2010 8:28	4.25
590306	22853	HAND WARMER URWEN JACK	6	12/1/2010 8:28	1.89
590306	22852	HAND WARMER RED POLKA DOT	6	12/1/2010 8:28	1.89
590307	84879	ASSORTED COLOUR BIRD ORNAMENT	52	12/1/2010 8:34	1.89
590307	22745	POPPY'S PLAYHOUSE BEDROOM	6	12/1/2010 8:34	2.3
590307	22748	POPPY'S PLAYHOUSE KITCHEN	6	12/1/2010 8:34	2.3
590307	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	12/1/2010 8:34	5.79
590307	22810	IVORY KNITTED MAUI COZY	6	12/1/2010 8:34	1.89
590307	84869	BOX OF 6 ASSORTED COLOUR TEMPOO...	6	12/1/2010 8:34	6.25
590307	22829	BOX OF VINTAGE JIGSAW BLOCKS	2	12/1/2010 8:34	4.95
590307	22832	BOX OF VINTAGE ALPHABET BLOCKS	2	12/1/2010 8:34	9.95
590307	21744	HOME BUILDING BLOCK WORDS	8	12/1/2010 8:34	5.99
590307	21755	LOVE BUILDING BLOCK WORDS	8	12/1/2010 8:34	5.95
590307	21777	SCOPS BOB WITH METAL HEART	4	12/1/2010 8:34	7.95
590307	48047	DOORMAT NEW ENGLAND	4	12/1/2010 8:34	7.95
590308	23960	LARK NAWING SET WITH LARKS	6	12/1/2010 8:34	4.25
590308	22813	RED COAT RACK PARIS FASHION	3	12/1/2010 8:34	4.95
590308	22812	YELLOW COAT RACK PARIS FASHION	3	12/1/2010 8:34	4.95
590308	22914	BLUE COAT RACK PARIS FASHION	3	12/1/2010 8:34	4.95
590309	21756	BATH BUILDING BLOCK WORDS	8	12/1/2010 8:35	5.95
590370	22728	ALARM CLOCK BAKELIKE PINK	24	12/1/2010 8:45	3.75
590370	22727	ALARM CLOCK BAKELIKE RED	24	12/1/2010 8:45	3.75
590370	22726	ALARM CLOCK BAKELIKE GREEN	12	12/1/2010 8:45	3.75
590370	21724	PANDA AND BUNNIES STICKER SHEET	12	12/1/2010 8:45	0.85
590370	23885	STARS GIFT TARE	24	12/1/2010 8:45	0.85
590370	20952	INFLATABLE POLITICAL GLOBE	48	12/1/2010 8:45	0.85

Algorithm 2

Data Transformation using Power Query

Step 1 Select Transform Data

Step 2 Open Power Query Editor

Step 3 Remove duplicates

Step 4 Handle null values

Step 5 Rename columns

Step 6 Apply changes

🔍 **Step 1: Check Columns:** Verify the imported columns (InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country).

🔍 **Step 2: Remove Blank Rows & Duplicates:** Select **Home** → **Remove Rows** → **Remove Blank Rows**, then select all columns (Ctrl + A) and choose **Remove Rows** → **Remove Duplicates**.

🔍 **Step 3: Standardize Text Values:** Right-click the Country column, select **Transform** → **Capitalize Each Word**.

🔍 **Step 4: Handle Missing Values:**

- Right-click CustomerID → **Replace Values** → Replace null with 0 (or Unknown Customer).
- Right-click Description → **Replace Values** → Replace null with Unknown Product.

🔍 **Step 5: Filter Invalid Values:** Click the filter dropdown on Quantity → **Number Filters** → **Greater Than** → 0. Repeat for UnitPrice (> 0).

🔍 **Step 6: Add Custom Column (TotalSales):** Go to **Add Column** → **Custom Column**, name it TotalSales, and enter the formula:

Code snippet

[Quantity] * [UnitPrice]

🔍 **Step 7: Set Data Types:**

- InvoiceNo, StockCode, Description, Country → **Text**
- InvoiceDate → **Date/Time** (or **Date**)
- CustomerID, Quantity → **Whole Number**
- UnitPrice, TotalSales → **Decimal Number**

🔍 **Step 8: Apply Changes:** Click **Home** → **Close & Apply** to load the cleaned dataset into Power BI.

Filter Rows

Apply one or more filter conditions to the rows in this table.

☒ Basic ☐ Advanced

Keep rows where 'Quantity'

is greater than or equal to

1

☒ And ☐ Or

Enter or select a valu...

OK

Cancel

Table with 7 columns: StockCode, Description, Quantity, UnitPrice, UnitCost, CustomerID, Country. The table displays a list of products and their associated data. The 'Quantity' column is highlighted in blue.

StockCode	Description	Quantity	UnitPrice	UnitCost	CustomerID	Country
811234	WHITE HANGING HEART T-LIGHT HOLDER	6	35.23	2.18	17859	United Kingdom
22883	HAND WARMER UNDER GLASS	6	35.23	2.18	17859	United Kingdom
84879	ASSORTED COLOUR BIRD ORNAMENT	52	35.23	2.18	17859	United Kingdom
22960	JAM MAKING SET WITH JARS	6	35.23	4.25	17859	United Kingdom
22756	BATH BUILDING BLOCK WORDS	3	35.23	5.80	17859	United Kingdom
22726	ALARM CLOCK SHARPER PINK	26	35.23	5.75	17859	United Kingdom
22086	PAPER CHAIN KIT SET CHRISTMAS	60	35.23	2.15	17859	United Kingdom
22852	HAND WARMER RED POLKA DOT	6	35.23	2.18	17859	United Kingdom
811234	WHITE HANGING HEART T-LIGHT HOLDER	6	35.23	2.18	17859	United Kingdom
22756	VETERAN SWING BOX LARGE	52	35.23	10.80	17859	United Kingdom
811234	WHITE HANGING HEART T-LIGHT HOLDER	6	35.23	2.18	17859	United Kingdom
22756	HOT WATER BOTTLE TEA AND EMPATHY	48	35.23	5.80	17859	United Kingdom
22852	HAND WARMER RED POLKA DOT	6	35.23	2.18	17859	United Kingdom
22786	JAMMED BAG PINK POLKA DOT	30	35.23	2.18	17859	United Kingdom
22961	JAM MAKING SET PRINTED	24	35.23	2.18	17859	United Kingdom
22756	REPRODUCED TEA SET COUNTRY 11 PC	24	35.23	4.25	17859	United Kingdom
0	Discount	1	35.23	27.8	17859	United Kingdom
10002	REPLACEMENT POLITICAL GLOBE	12	35.23	8.80	17859	United Kingdom
350942	SET OF 3 COLOURED FLYING DUCKS	1	35.23	4.80	17859	United Kingdom
82484	WOOD BLACK BOARD AND WHITE PEN	3	35.23	6.40	17859	United Kingdom
22786	SET 3 WOODEN BASKETS W LIDS	3	35.23	10.80	17859	United Kingdom
84880	WHITE WIRE EGG HOLDER	48	35.23	6.80	17859	United Kingdom
79023	CHALK LIGHTS	262	35.23	5.80	17859	United Kingdom
22756	HOT WATER BOTTLE BLOCK WORDS	3	35.23	5.80	17859	United Kingdom
22941	CHRISTMAS LIGHTS 10 REMINDERS	6	35.23	8.1	17859	Australia
22941	CHRISTMAS LIGHTS 10 REMINDERS	2	35.23	8.1	17859	United Kingdom
22786	PLAYING IN THE CIRCLE PARKER	12	35.23	5.80	17859	United Kingdom
22786	2 BERRY WIRE RECLAMANT	6	35.23	2.18	17859	United Kingdom

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22756	HOT WATER BOTTLE TEA AND EMPATHY	48	35.23	5.80	17859	United Kingdom
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22786	JAMMED BAG PINK POLKA DOT	30	35.23	2.18	17859	United Kingdom
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82484	WOOD BLACK BOARD AND WHITE PEN	3	35.23	6.40	17859	United Kingdom
22786	SET 3 WOODEN BASKETS W LIDS	3	35.23	10.80	17859	United Kingdom
84880	WHITE WIRE EGG HOLDER	48	35.23	6.80	17859	United Kingdom
79023	CHALK LIGHTS	262	35.23	5.80	17859	United Kingdom
22756	HOT WATER BOTTLE BLOCK WORDS	3	35.23	5.80	17859	United Kingdom
22941	CHRISTMAS LIGHTS 10 REMINDERS	6	35.23	8.1	17859	Australia
22941	CHRISTMAS LIGHTS 10 REMINDERS	2	35.23	8.1	17859	United Kingdom
22786	PLAYING IN THE CIRCLE PARKER	12	35.23	5.80	17859	United Kingdom
22786	2 BERRY WIRE RECLAMANT	6	35.23	2.18	17859	United Kingdom

Table.TransformColumns({Removed Blank Rows},{Country, Text.Proper, type text})							
	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
1	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 09:26:00	2.55	17850	United Kingdom
2	12633	HAND WARMER UNION JACK	6	01-12-2010 09:28:00	1.85	17850	United Kingdom
3	84879	ASSORTED COLOUR BIRD ORNAMENT	32	01-12-2010 09:34:00	1.69	13047	United Kingdom
4	12960	JAM MAKING SET WITH JARS	6	01-12-2010 09:34:00	4.25	13047	United Kingdom
5	12156	BATH BUILDING BLOCK WORD	3	01-12-2010 09:35:00	5.95	13047	United Kingdom
6	22728	ALARM CLOCK-BABELE PINK	24	01-12-2010 09:45:00	5.75	12585	France
7	12089	PAPER CHAIN KIT 50'S CHRISTMAS	80	01-12-2010 09:00:00	2.55	15748	United Kingdom
8	12632	HAND WARMER RED POLKA DOT	6	01-12-2010 09:01:00	1.85	17850	United Kingdom
9	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 09:02:00	2.55	17850	United Kingdom
10	12158	VICTORIAN SEWING BOX LARGE	52	01-12-2010 09:09:00	10.95	15109	United Kingdom
11	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 09:12:00	2.55	17850	United Kingdom
12	22134	HOT WATER BOTTLE TEA AND SYMPATHY	48	01-12-2010 09:32:00	8.45	15291	United Kingdom
13	12632	HAND WARMER RED POLKA DOT	6	01-12-2010 09:34:00	1.85	17850	United Kingdom
14	22386	JUMBO BAG PINK POLKA DOT	20	01-12-2010 09:57:00	1.95	14688	United Kingdom
15	12961	JAM MAKING SET PRINTED	24	01-12-2010 09:41:00	1.45	17809	United Kingdom
16	12138	RETROSPOT TEA SET CERAMIC 15 PC	20	01-12-2010 09:41:00	4.25	15111	United Kingdom
17	0	Discount	-1	01-12-2010 09:41:00	27.5	14527	United Kingdom
18	10902	INFLATABLE POLITICAL GLOBE	12	01-12-2010 09:45:00	0.83	16098	United Kingdom
19	1604C	SET OF 5 COLOURED FLYING DUCKS	-1	01-12-2010 09:48:00	4.83	15111	United Kingdom
20	82484	WOOD BLACK BOARD ART WHITE FINISH	3	01-12-2010 09:52:00	6.45	18074	United Kingdom
21	22783	SET 3 WICKER OVAL BASKETS W UDS	1	01-12-2010 09:56:00	10.95	17420	United Kingdom
22	84880	WHITE WIRE EGG HOLDER	36	01-12-2010 09:57:00	4.95	16029	United Kingdom
23	78521	CHILL LIGHTS	100	01-12-2010 09:58:00	5.82	16029	United Kingdom
24	12154	HOME BUILDING BLOCK WORD	3	01-12-2010 09:59:00	5.95	16250	United Kingdom
25	12941	CHRISTMAS LIGHTS 25 REINFORCED	8	01-12-2010 10:01:00	8.5	12451	Australia
26	12941	CHRISTMAS LIGHTS 25 REINFORCED	2	01-12-2010 10:23:00	8.5	17511	United Kingdom
27	12158	PLASTER IN PIN CIRCUS PARADE	-12	01-12-2010 10:24:00	1.65	17549	United Kingdom
28	12150	3 STRIPEY ACE FELTCRAFT	6	01-12-2010 10:25:00	1.95	17705	United Kingdom

Algorithm 3: Creating DAX Measures & KPI Cards

- **Step 1:** In Report View, navigate to **Table Tools / Measure Tools** and click **New Measure**:
- **Maximum Sales Measure:**

MaximumSales = MAX('OnlineRetail'[TotalSales])

MaximumSales

998.4

- **Minimum Sales Measure:**

```
MinimumSales = MIN('OnlineRetail'[TotalSales])
```

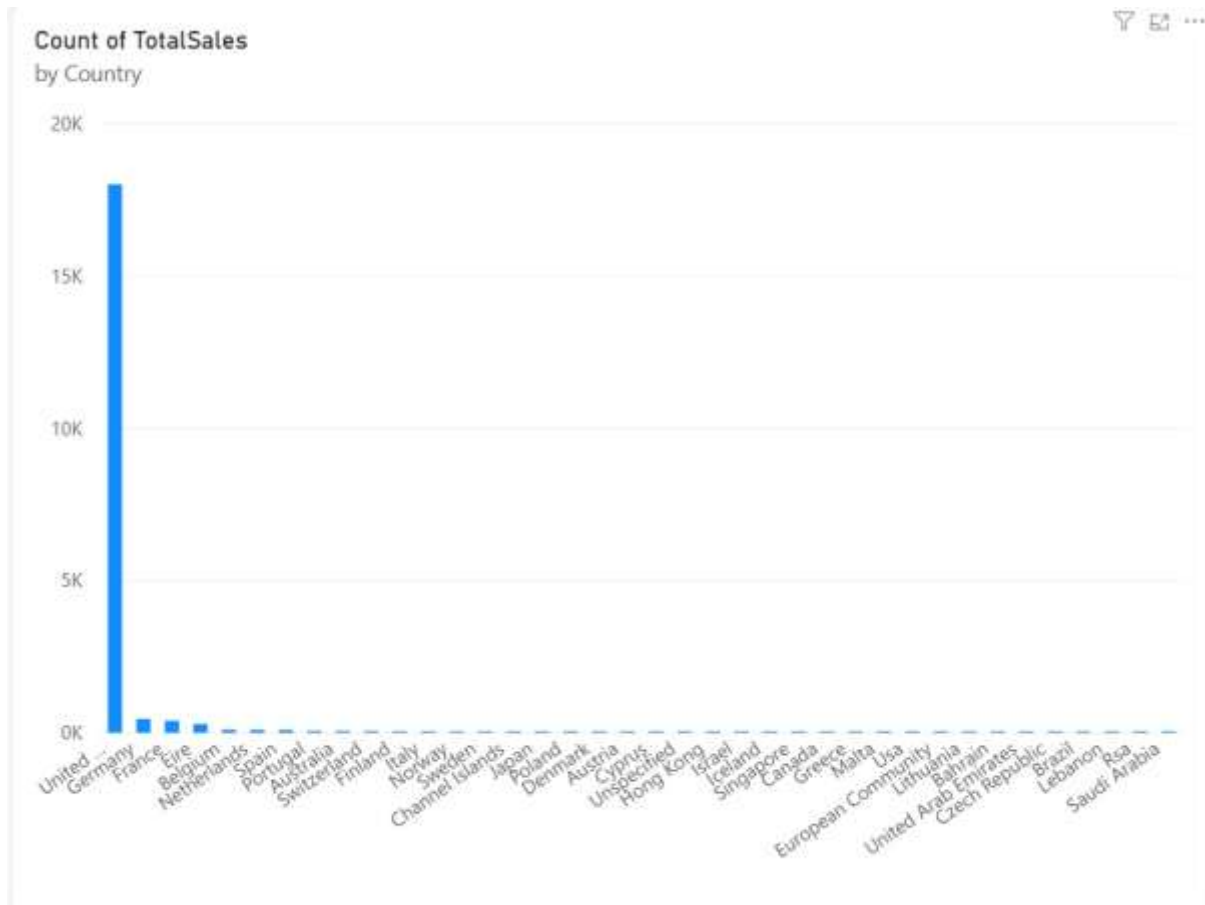
MinimumSales

0.1

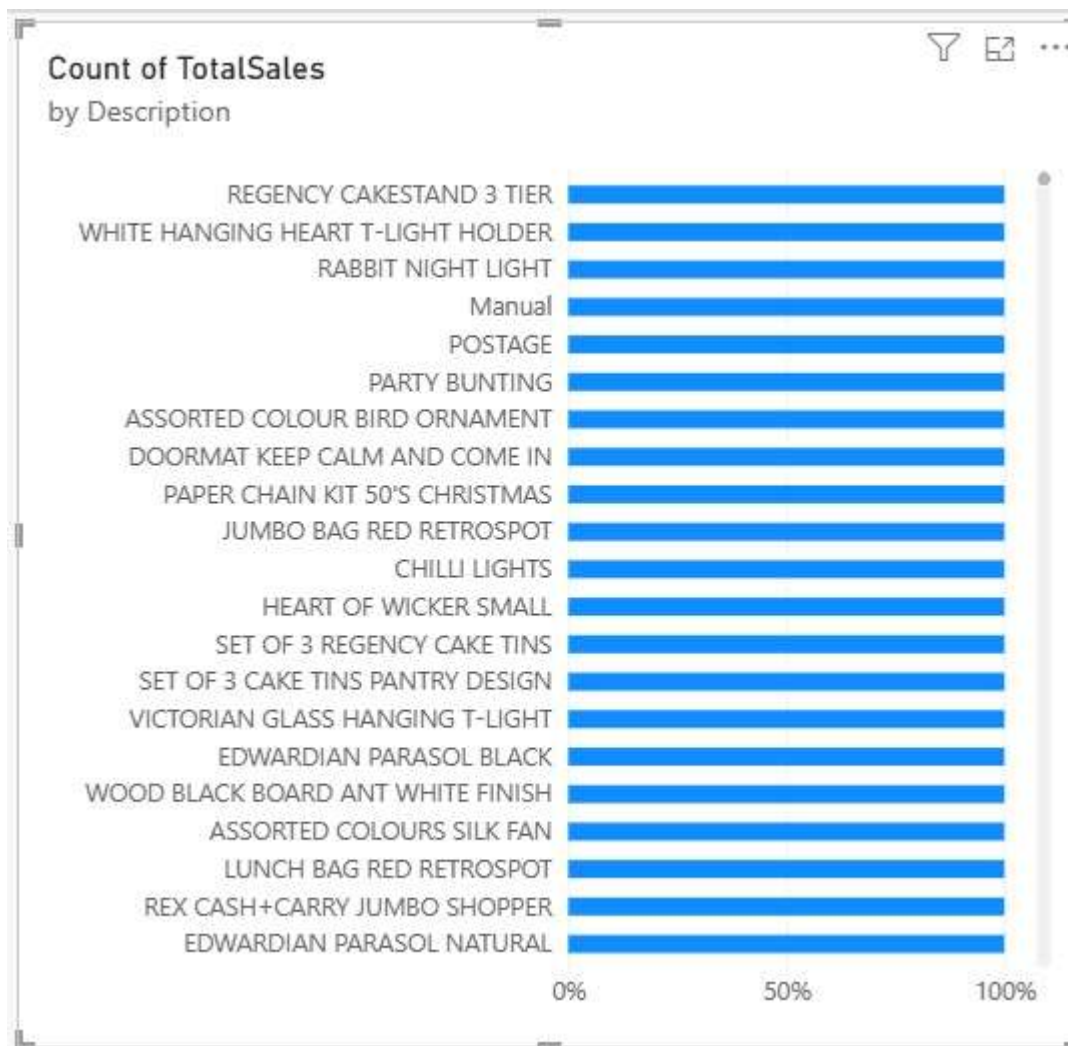
- **Step 2:** Select the **Card** visual from the Visualizations pane.
- **Step 3:** Drag each measure into separate card visual fields to display overall KPIs.

Algorithm 4: Creating Charts & Visualizations

- **Clustered Column Chart (Sales by Country):**
 - Select **Clustered Column Chart**.
 - Drag Country to the **X-axis** and TotalSalesMeasure (or TotalSales) to the **Y-axis**.

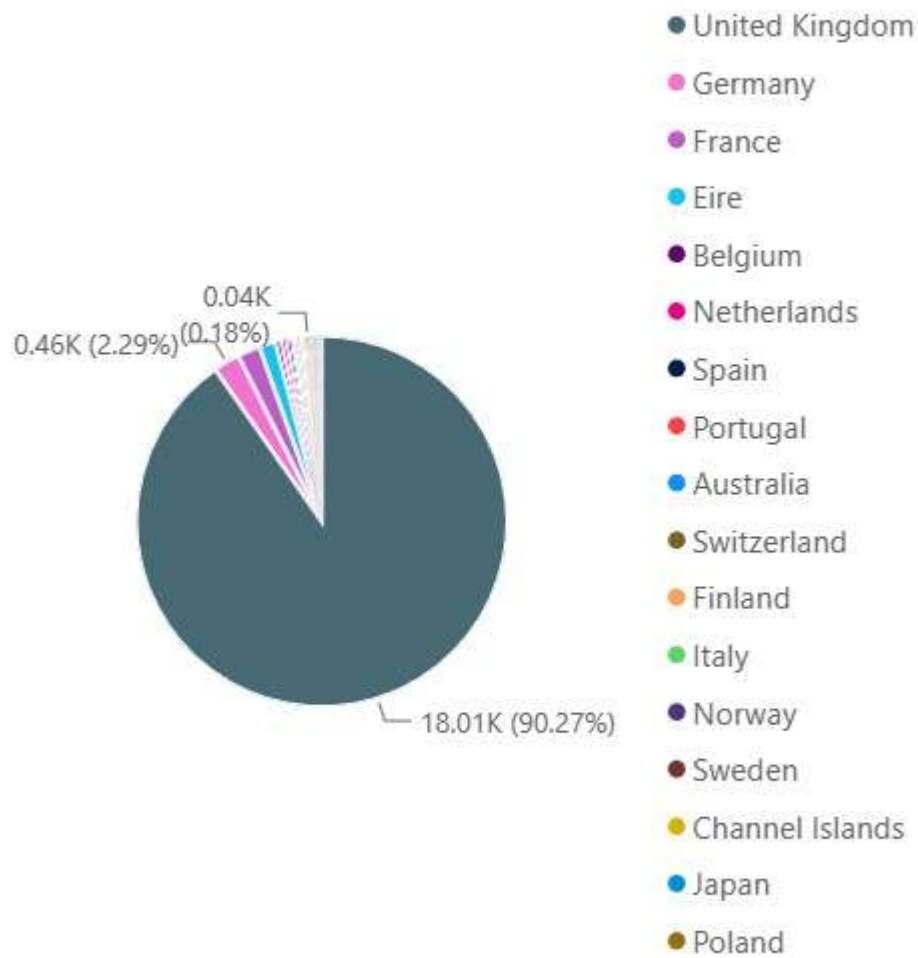


- **Bar Chart (Top Products by Sales):**
 - Select **Stacked Bar Chart**.
 - Drag Description to the **Y-axis** and TotalSales to the **X-axis**.

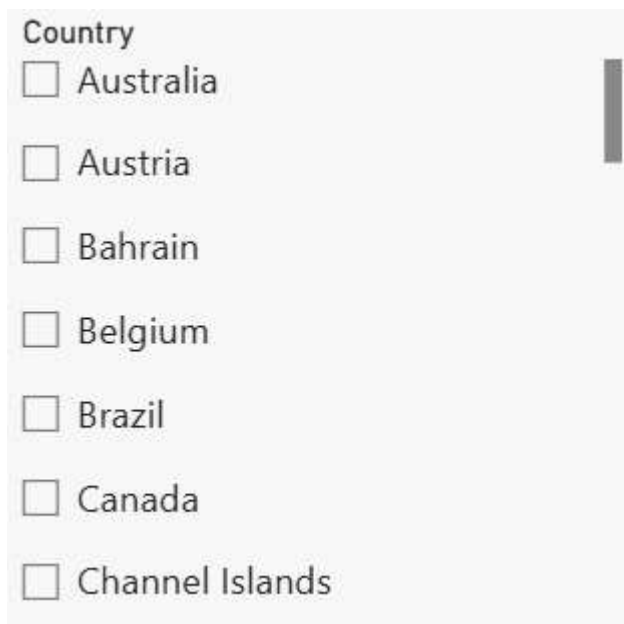


- **Pie / Donut Chart (Sales Contribution):**
 - Select **Pie Chart**.
 - Drag Country to **Legend** and TotalSales to **Values**.

Count of TotalSales by Country

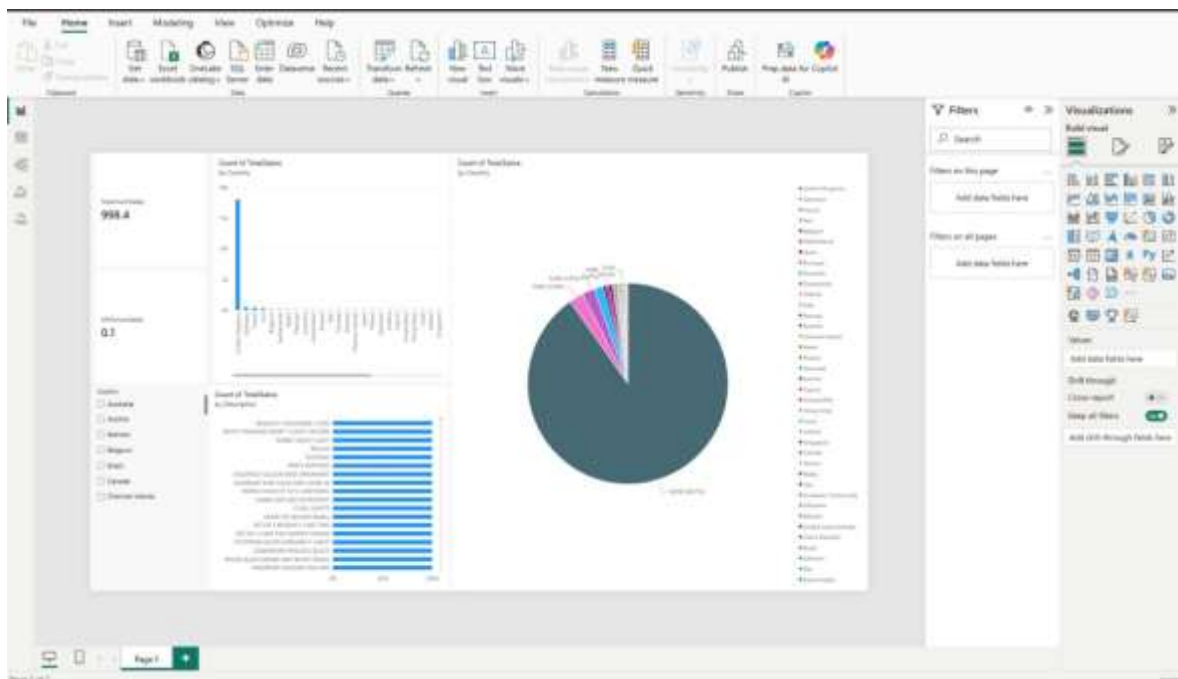


- **Interactive Slicer:**
 - Select the **Slicer** visual.
 - Drag Country into the **Field** well.



Algorithm 5: Building the Dashboard

- **Step 1:** Arrange the 4 KPI Cards at the top/side.
- **Step 2:** Position the Bar Chart, Column Chart, and Pie Chart on the canvas.
- **Step 3:** Place the Country Slicer for interactive filtering.
- **Step 4:** Format visual titles, data labels, and colors, then save the report (.pbix).



Conclusion: The dataset was cleaned, transformed, and modeled in Power BI Desktop to build an interactive retail sales dashboard for business performance tracking